Alper Demir LinkedIn | Google Scholar | ORCID | Web of Science | Scopus

Contact Information
İzmir Institute of Technology,
Department of Computer Engineering, #141
https://gliese876b.github.io/
alperdemir@iyte.edu.tr

Research Interests
Machine Learning, Reinforcement Learning, Multi-Agent Reinforcement Learning

Education

Middle East Technical University (METU), Ankara, Turkey

Ph.D., Computer Engineering, November 2019

- Topic: *Improving Reinforcement Learning using Distinctive Clues from the Environment*
- CGPA: 3.93 / 4.00
- Advisor: Prof. Dr. Faruk Polat , Co-Advisor: Dr. Erkin Çilden

M.Sc., Computer Engineering, August 2016

- GPA: 3.79 / 4.00
- Topic: *Effective Subgoal Discovery and Option Generation in Reinforcement Learning*
- Advisor: Prof. Dr. Faruk Polat

B.Sc., Computer Engineering, June 2014

- GPA: 3.36 / 4.00, HONOR 7 Semesters, HIGH HONOR 1 Semester
- Erasmus Program : Wrocław Uni. of Technology, Wrocław, Poland, 2013

Academic Experience

Assistant Prof. August 2026 - ...
Department of Computer Engineering, Izmir Institute of Technology

Assistant Prof. February 2020 - July 2026
Department of Computer Engineering, Izmir University of Economics

Postdoctoral Researcher August 2024 - August 2025
Autonomous Agents Research Group, University of Edinburgh

Research & Teaching Assistant September 2014 - November 2019
Department of Computer Engineering, Middle East Technical University
Supervisor: Prof. Dr. Faruk Polat, Co-Supervisor: Dr. Erkin Çilden

Industry Experience

Software Developer, BTT Inc., METU Technopolis 2013-2014
- Developing applications on Android
- Involving in Web Development

Intern, Turkish Aerospace Industries (TAI) Summer 2012
- Working with senior developers in designing and developing software tools
- Writing tests for Human Machine Interface partition

Academic Service

Conference Reviewing
- Association for the Advancement of Artificial Intelligence (2022)
- European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (2021, 2022)
- International Informatics and Software Engineering Conference (2022)
- Multi-disciplinary Conference on Reinforcement Learning and Decision Making (2025)
- International Conference on Emerging Trends and Applications in Artificial Intelligence (2026)

Journal Reviewing
- Applied Intelligence (2023, 2024)
- Artificial Intelligence Review (2024)
- International Journal of Machine Learning and Cybernetics (2024)
- Artificial Intelligence Communications (2024)
- Knowledge-based Systems (2024)
- Machine Learning (2024)

Projects

- Academic Advisor, *Developing an AI-Backed Personal Management System Based on Real-Time Data for Music Artists*,
  University-Industry Collaboration Project,
  2025 - 2026, İzmir, Turkey
- Academic Advisor, *Development of an Artificial Intelligence Infrastructure for Scoring Digital Music Works and Discovering New Music Pieces with High Success Potential*,
  University-Industry Collaboration Project,
  2024 - 2025, İzmir, Turkey
- Principal Investigator, *Solving Asymmetric Social Dilemmas in Multi-Agent Reinforcement Learning with Local Information*,
  Scientific and Technological Research Council of Turkey (TÜBİTAK),
  International Postdoctoral Research Fellowship Program for Turkish Citizens, 2024 - 2025, Edinburgh, United Kingdom
- Academic Advisor, *APP2SCALE – AI-Based Auto-Scaling and Fine-Tuning*,
  University-Industry Collaboration Project,
  2023 - 2024, İzmir, Turkey
- Principal Investigator, *Forming Useful Memory in Partially Observable Reinforcement Learning*,
  Scientific and Technological Research Council of Turkey (TÜBİTAK),
  Grant No: 120E427, 2021 - 2023, İzmir, Turkey
- Research Fellow, *YÖK Program Ve Kontenjan Yönetim Sistemi (YÖK PKYS)*,
  Scientific and Technological Research Council of Turkey (TÜBİTAK),
  Grant No: 115G086, 2018 - 2019, Ankara, Turkey
- Research Fellow, *Subgoal Identification in Sequential Decision Making under Partial Observability*,
  Scientific and Technological Research Council of Turkey (TÜBİTAK),
  Grant No: 215E250, 2016 - 2018, Ankara, Turkey

Journal Publications

1. Demir, A., Çilden, E. & Polat, F. Generating Effective Initiation Sets for Subgoal-Driven Options. *Advances in Complex Systems* **22.** doi:10.1142/S0219525919500012 (2019).
2. Demir, A., Çilden, E. & Polat, F. Automatic landmark discovery for learning agents under partial observability. *The Knowledge Engineering Review* **34.** doi:10.1017/S026988891900002X (2019).
3. Demir, A., Çilden, E. & Polat, F. Landmark based guidance for reinforcement learning agents under partial observability. *International Journal of Machine Learning and Cybernetics,* 1–21 (2022).
4. Demir, A. Learning what to memorize: Using intrinsic motivation to form useful memory in partially observable reinforcement learning. *Applied Intelligence,* 1–19 (2023).
5. Uner, B. Y., Demir, A., Zhou, P., Taskiran, E. Z. & Wassenaar, T. Peptide-nanoparticle platforms for antisense therapeutics: A coarse-grained modeling approach to brain delivery. *Computers in Biology and Medicine* **203,** 111479. ISSN: 0010-4825 (2026).
6. Eren, B. & Demir, A. Effective Exploration via Intrinsic Motivation in Reinforcement Learning. *Journal of Advanced Research in Natural and Applied Sciences* **12,** 171–192.

Conference Proceedings

7. Demir, A., Çilden, E. & Polat, F. *A history tree heuristic to generate better initiation sets for options in reinforcement learning* in *Proceedings of the Twenty-second European Conference on Artificial Intelligence (ECAI)* (2016), 1644–1645. doi:10.3233/978-1-61499-672-9-1644.
8. Demir, A., Çilden, E. & Polat, F. *Local roots: A tree-based subgoal discovery method to accelerate reinforcement learning* in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECML/PKDD)* (2016), 361–376. doi:10.1007/978-3-319-46227-1_23.
9. Demir, A., Çilden, E. & Polat, F. *A Concept Filtering Approach for Diverse Density to Discover Subgoals in Reinforcement Learning* in *2017 IEEE 29th International Conference on Tools with Artificial Intelligence (ICTAI)* (2017), 1–5. doi:10.1109/ICTAI.2017.00012.

10. Demir, A., Çilden, E. & Polat, F. *Landmark Based Reward Shaping in Reinforcement Learning with Hidden States* in *Proceedings of the 18th International Conference on Auto-nomous Agents and MultiAgent Systems* (Int. Foundation for Autonomous Agents and Multiagent Systems, 2019), 1922–1924. ISBN: 978-1-4503-6309-9.
11. Dilek, İ., Oğuz, K. & Demir, A. *Behaviour Analysis of Izmir Residents Using Public Wi-Fi Access Point Usage* in *2022 3rd International Informatics and Software Engineering Conference (IISEC)* (2022), 1–6. doi:10.1109/IISEC56263.2022.9998278.
12. Başoğul, A. O., Çekirdek, H., Nakipoğlu, K., Yağcı, S. & Demir, A. *Analyzing Traffic Patterns in Izmir: A Study on Busy Hours and Congestion* in *2024 Innovations in Intelligent Systems and Applications Conference (ASYU)* (2024), 1–6. doi:10.1109/ASYU62119.2024.10757054.
13. Arslan, A., Titiz, İ. E., Gürcan, E. C. & Demir, A. *Water Consumption Dynamics in Izmir: Analyzing Influences of District, Seasonality, and External Events* in *2024 Innovations in Intelligent Systems and Applications Conference (ASYU)* (2024), 1–6. doi:10.1109/ASYU62119.2024.10757161.
14. Çakır, M. T., Nayir, H., Demir, A., Kaya, H. & Ceylan, O. *A reinforcement learning based approach to solve voltage issues in distribution networks* in *2025 IEEE 19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG)* (2025), 1–6. doi:10.1109/CPE-POWERENG63314.2025.11027244.
15. Nayir, H., Demir, A. & Kaya, H. *Multi-Dimensional Autoscaling in Microservices Using Rein forcement Learning* in *2025 33rd International Symposium on Modeling, Analysis and Sim. of Computer and Telecom. Sys. (MASCOTS)* (2025), 1–8. doi:10.1109/MASCOTS67699.2025.11283269.
16. Demirbilek, B. H., Demir, A. & Polat, F. *External Visual Memory with Autoencoder-Based Intrinsic Motivation for Reinforcement Learning Under Partial Observability* in *2025 IEEE 37th International Conference on Tools with Artificial Intelligence (ICTAI)* (2025), 796–800. doi:10.1109/ICTAI66417.2025.00115.
17. Demir, A., Kaya, H., Uluğ, M. & Özkan, I. *Reinforcement Learning for Autoscaling in Cloud Environments: A Case Study from Industry* in *2025 16th International Conference on Electrical and Electronics Engineering (ELECO)* (2025), 1–5. doi:10.1109/ELECO69582.2025.11329259.
18. Demir, A., Aydın, H., Tessera, K.-a., Abel, D. & Albrecht, S. V. *Fairness over Equality: Correcting Social Incentives in Asymmetric Sequential Social Dilemmas* in *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems* (2026), 13–21. doi:10.65109/CZPZ7833.
19. Uz, E. K., Akgün, H. A., Yontar, H. & Demir, A. *Spatiotemporal Analysis and Incident Severity Forecasting: Unveiling Traffic Patterns in Metropolitan Izmir* in *2026 8th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA)* (2026), 1–9. doi:10.1109/ICHORA69329.2026.11537079.
20. Tur, B., Kobak, E. A., Koğuş, M. & Demir, A. *Temporal and Spatial Dynamics of Public Transit in Istanbul: A Data-Driven Analysis of Passenger Demand Patterns* in *2026 8th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA)* (2026), 1–8. doi:10.1109/ICHORA69329.2026.11537204.

References

Prof. Dr. Faruk Polat, polat@ceng.metu.edu.tr
Department of Computer Engineering, Middle East Technical University
Ankara, Turkey

Dr. Erkin Çilden, erkin.cilden@stm.com.tr
STM Defense Technologies Engineering and Trade Inc.
Ankara, Turkey

Prof. Dr. İsmail Hakkı Toroslu, toroslu@ceng.metu.edu.tr
Department of Computer Engineering, Middle East Technical University
Ankara, Turkey

Assoc. Prof. Dr. Yusuf Sahillioğlu, ys@ceng.metu.edu.tr
Department of Computer Engineering, Middle East Technical University
Ankara, Turkey

Assoc. Prof. Dr. Sinan Kalkan, skalkan@ceng.metu.edu.tr
Department of Computer Engineering, Middle East Technical University
Ankara, Turkey